THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Assignment 4, 2025

School of Mathematics and Statistics

MAST20009 Vector Calculus

Submission deadline: 5pm on Monday May 26

This assignment consists of 11 pages (including this page) with 4 questions and 44 total marks

Instructions to Students

- This assignment is worth 5% of your total MAST20009 mark.
- If you have a printer, print the assignment one-sided.

Writing

- Assignments must be neatly written in blue or black pen or may be typeset using LaTeX.
- Full working must be shown in your solutions.
- Marks will be deducted for incomplete working, insufficient justification of steps, incorrect mathematical notation and for messy presentation of solutions.
- If you are writing answers on the assignment and you need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

Scanning and Submitting

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned assignment as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary.

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Question 1 (8 marks)

Let $P = ABCD$ be the parallelogram in the xy-plane with

$$A = (0,0), \quad B = (-2,1), \quad C = (-1,4), \quad D = (1,3).$$

Let $P' = A'B'C'D'$ be the square in the uv-plane with

$$A' = (0,0), \quad B' = (1,0), \quad C' = (1,1), \quad D' = (0,1).$$

(a) Find a bijective linear map which takes P' to P, such that $A' \mapsto A$, $B' \mapsto B$, $C' \mapsto C$, and $D' \mapsto D$.

(b) Evaluate the double integral

$$\iint_P (xy + x^2)\, dx\, dy.$$

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Question 2 (10 marks)

Let S be the ellipsoid defined by

$$4x^2 + 4y^2 + z^2 = 4.$$

Write down a double integral which gives the surface area of that part of S which is above the xy-plane. The integrand should be an explicit scalar function, i.e. not written in terms of a cross product. *You do not need to evaluate the integral.*

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Question 3 (14 marks)

Let S_1 be the surface of the cone $z = \sqrt{x^2 + y^2}$, $0 \leq z \leq 1$, and let S_2 be the round "lid" at the top of S_1. Let $S = S_1 \cup S_2$, so that S is a closed surface. Orient S so that its normal vectors are in the outward direction.

(a) Find a parametrisation $\mathbf{T}$ of S_1 and write down its normal vector.

(b) Let $\mathbf{F}$ be the vector field $\mathbf{F}(x, y, z) = (y, 0, z)$. Compute the flux

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS,$$

where $\hat{\mathbf{n}}$ is the outward normal to the surface S.

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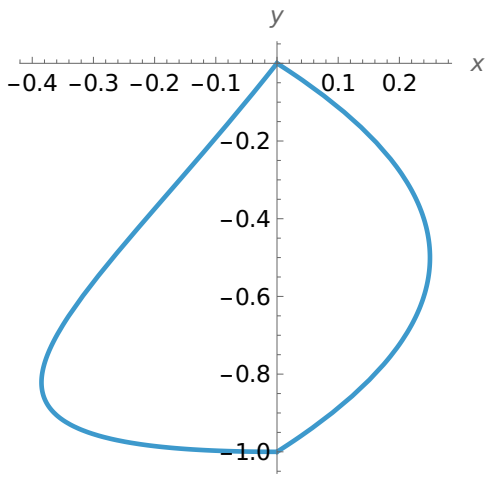
Question 4 (12 marks)

Consider the curve C in $\mathbb{R}^2$ with parametrisation

$$\mathbf{c}(t) = \begin{cases} (t^3 - t,\, t^2 - 2t), & 0 \le t < 1 \\ (-t^2 + 3t - 2,\, t - 2), & 1 \le t \le 2. \end{cases}$$

It looks like this:



(a) Prove that with the orientation given by increasing t, C is positively oriented.

(b) Find the area of the region D inside of C.

End of Assignment — Total Available Marks = 44

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